

Whole body protection against lethal ionizing radiation in mice by REC-2001: A semi-purified fraction of *Podophyllum hexandrum*

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Abstract

The current study has concentrated on assessment of the radioprotective potential of REC-2001, a semi-purified fraction of rhizomes of *Podophyllum hexandrum*, in Swiss albino Strain 'A' mice exposed to 10 Gy whole-body gamma radiation. Animals were treated with 10 and 15 mg/kg bwt (i.p.) of REC-2001 1 h prior to exposure to a lethal dose of γ -radiation (10 Gy) and observed upto 30 days. For analysis of maximum tolerable dose (MTD), LD₅₀ and acute toxic dose, different concentrations of the extract were administered to animals and their mortality and morbidity status was observed upto 72 h and one week, respectively. Dose reduction factor (DRF) was determined by exposing REC-2001 pre-treated mice to supra-lethal doses of γ -radiation. Endogenous spleen colony forming units (CFU), DNA strand breaks in thymocytes (alkaline halo assay) and lipid degradation was studied to understand the mechanism of radioprotection. A single dose of REC-2001 (10 and 15 mg/kg bwt i.p.) exhibited >90% survival in the pre-treated irradiated group versus no survival in radiation control group. Single doses of upto 75 mg/kg bwt (i.p.) did not cause any mortality (MTD) in mice. REC-2001, a dose of 90 mg/kg bwt, resulted in 50% mortality (LD₅₀), while the LD₁₀₀ was 115 mg/kg bwt REC-2001 exhibited a DRF of 1.62. CFU counts in the REC-2001 treated group were found significantly high (5.33/spleen) as compared to controls. Exposure of thymocytes to 10 Gy radiation resulted in increased halo diameter ($45 \pm 3 \mu\text{m}$) in comparison to untreated controls ($8 \pm 1 \mu\text{m}$). REC-2001 administration (500 $\mu\text{g}/\text{ml}$) decreased the halo diameter to $15 \pm 2 \mu\text{m}$. Radiation-induced lipid degradation was also inhibited by REC-2001. The present study has revealed that REC-2001 is a promising radioprotective fraction that can be effectively used against lethal doses of γ -radiation after further investigations in higher animal models.

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Introduction

A wide range of synthetic (Maisin, 1998; Giambarresi and Jacobs, 1987) as well as natural compounds (Weiss and Landauer, 2003) have been tested in experimental models to overcome the deleterious effects inflicted by ionizing radiation. However, the use of presently

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available synthetic radio protective agents has been limited due to their inherent toxic nature, particularly at therapeutically effective concentrations (Bump and Malaker, 1998; Sweeney, 1979). Since radiation protection is a multi factorial phenomenon and is mediated via a number of mechanisms such as scavenging of free radicals, metal chelation, inhibition of lipid peroxidation, etc., therefore, use of single compounds to minimize radiation-induced injuries may not yield the desired results. In view of this, exploration for newer and effective agents has been continuing. Phytoextracts are reported to contain a large number of bio-active molecules, each with different properties, which in synergism render protection to most of the vital organs of living systems against various stresses (Shimoi et al., 1994; Duh and Yen, 1997; Uma Devi et al., 1999).

In recent years, a number of plants have been screened for their radioprotective potential in both in vitro and in vivo models (Arora et al., 2005; Jagetia and Aruna, 1997; Goel et al., 2002; Uma Devi et al., 1999). *Podophyllum hexandrum* Royale syn. *P. emodi* Wall (Himalayan May apple; Family: Berberidaceae), a perennial rhizomatous herb that inhabits the Himalayan regions, has been extensively used in *Ayurveda* and traditional Chinese medicine for treatment of ailments like neurological disorders (Blasko and Cordell, 1998), cancer of various organs (Singh and Shah, 1994; Blasko and Cordell, 1998) venereal warts (Beutner, 1996) leukemia, Hodgkin's and non-Hodgkin's lymphoma, (Singh and Shah, 1994), viral and bacterial infections (Gowdey et al., 1995) etc. The rhizomes of *P. hexandrum* are reported to contain lignans such as podophyllotoxin, 4-demethylpodophyllotoxin, podophyllotoxone, podophyllotoxin glycoside, 4-demethylpodophyllotoxone, deoxypodophyllotoxin, dehydropodophyllotoxin, and other aryl tetrahydronaphthalene lignans (Singh and Shah, 1994; Wong et al., 2000). In addition, several flavonoids including quercetin, kaempferol, astragalin, and kaempferol -3-glucoside (Singh and Shah, 1994; Wong et al., 2000) also have been identified in *Podophyllum* sps. The radioprotective action of crude extract of *P. hexandrum* has been partially elucidated based on its various characteristics, i.e., scavenging free radical, chelating transition metals, reducing lipid degradation, modulation of antioxidant enzyme levels etc. (Kumar and Goel, 2000; Mittal et al., 2001). REC-2001, an alcoholic fraction of *P. hexandrum*, has been evaluated earlier for its chemical constituents (Sagar et al., 2006). This study, for the first time, reveals whole body protection ability of REC-2001 against lethal doses of gamma radiation in Swiss albino mice.

The present study reports the evaluation of REC-2001, in terms of its effect on whole body survival in lethally irradiated strain 'A' mice. Dose reduction factor (DRF) for REC-2001 was also analyzed in mice to observe its potential against supra lethal radiation.

Studies on endogenous spleen colony forming units, lipid peroxidation and DNA damage were performed in order to ascertain its protective effect at tissue and organ level.

Materials and methods

Collection of plant material

Podophyllum hexandrum Royale (syn. *Podophyllum emodi* wall) was collected from Sonmarg, (Jammu and Kashmir state, India) and transplanted under natural conditions at the RRL Bonera farm. Voucher specimens have been deposited at the repository of RRL, Srinagar (Voucher no: RRL/PH/Srinagar-2004).

Method of extraction

Extraction was performed as described in Sagar et al., 2006. Briefly, rhizome samples of *P. hexandrum* were washed thoroughly with running water and dried under partial shade. The dried samples were powdered and subjected to petroleum ether extraction. This was followed by extraction with 95% ethyl alcohol for 80 h. Podophyllin was precipitated using 1% HCl. The viscous extract was filtered and vacuum dried. The dried resin was refluxed in a Soxhlet apparatus (below 60 °C) utilizing petroleum ether, followed by extraction with chloroform. The chloroform soluble extract was dissolved in methanol and boiled with neutral alumina and the solution was filtered and evaporated under vacuum in a rotavapor (Buchi, Switzerland). The left over residue was dissolved in methanol, and benzene (2:1) was added to crystallize the podophyllotoxin. The mother liquor left after removal of podophyllotoxin was concentrated and designated as REC-2001.

HPLC analysis

Isocratic analysis

REC-2001 was analyzed on Shimadzu LC-10 AT/VP HPLC machine isocratically, utilizing E. Merck RP-18 e column (250 × 4.0 mm 5 μm) with diode array detector SPD M-10 A VP/RF-10 AXL fluorescent detector, and auto injector SIL-10 AD VP. Elution was done with the mobile phase (MeOH:H₂O; 60:40) for 25 min at a flow rate of 0.8 ml/min and a wavelength of 290 nm was used for measurement.

Gradient analysis

The compounds were analyzed on Shimadzu LC-10 AT/VP using a diode detector with mobile phase consisting of methanol (A) and water (B), 0.01–5 min (A:B; 65:35), 5–65 min (A:B; 35:65) and finally 65–75 min (A:B; 65:35),

E. Merck RP-18 e column (5 μ m, 4 $\times$ 250 mm). Separation was carried out at 30 °C with a flow rate of 0.6 ml/min. Measurements were taken at 290 nm. Characterization of active constituents was carried out by co-spiking with standard markers and by LC–MS and LC–MS/MS data as described earlier by Sagar et al. (2006). Lignans were quantified by standard calibration. The qualitative analysis of three flavinoids, viz. quercetin, kaempferol and kaempferol-3-glucoside was done by comparing their TLC, HPTLC and HPLC profiles with reference standard markers.

Experimental design

Swiss albino Strain ‘A’ male mice (8–10 weeks; weight 27 ± 2 g) were maintained under controlled laboratory conditions and all the experiments were conducted strictly in accordance with the guidelines of the Institutional Animal Ethics Committee (IAEC). The treated groups always had a minimum of 18–24 animals.

Drug administration

REC-2001 was dissolved in DMSO and diluted with distilled water so that the final concentration of DMSO was less than 25%. The doses (expressed in mg/kg body weight) refer to the weight of lyophilized preparation of REC-2001 administered per kg body weight. Different doses of REC-2001 were administered to mice intraperitoneally (i.p.). Control animals were injected with 0.25 ml of normal saline.

Maximum tolerable dose (MTD) and acute toxicity

The maximum concentration of REC-2001, which did not bring about death/severe toxic manifestations in the experimental animals, was considered as MTD. Acute toxicity of REC-2001 was determined in terms of percent survival of animals. Concentration of the dose resulting in 50% death was considered as LD₅₀.

Irradiation

Radiation was delivered to mice from a ⁶⁰Co gamma chamber (Model-220, Atomic Energy of Canada Ltd.) at a dose rate of 0.56 cGy/s during the experimental period. Dosimetry was carried out using Baldwin Farmer’s secondary dosimeter and Fricke’s chemical dosimetry method.

Dose reduction factor was calculated by the following method (Miller and Tainter, 1944):

$$\text{DRF} = \frac{(\text{LD}_{50} \text{ of REC} - 2001 + \text{Radiation group})}{\text{LD}_{50} \text{ of Radiation group}}.$$

Whole-body survival and body weight

The effect of variable time intervals between administration of different concentrations of REC-2001 and

irradiation on survival and body weight of mice was investigated. Survival was recorded daily up to 30 post-irradiation days, and data was expressed as percent survival. Mice used for the study were distributed into different groups, viz., control, irradiated, REC-2001 treated and REC-2001 plus radiation group. Each group consisted of 6 mice. Body weight of the animals was recorded every alternate day. Change in average body weight of animals at different time intervals was calculated considering the initial body weight of the animal as 100%.

Endogenous spleen colony forming unit (CFU) assay

For endogenous CFU assay, mice from each group were sacrificed by cervical dislocation on the 10th post-irradiation day; spleens were removed and fixed in Bouin’s solution for 24 h. Colony forming units (CFU), visible to naked eyes, were scored from each spleen.

Alkaline halo assay

DNA strand breaks in thymocytes were detected using the alkaline halo assay (Sestili et al., 1998). Measurements were done from center of the core to any point on the perimeter of the halo and at least 100 cells were counted from each treatment. DNA damage was expressed as % DNA damage in comparison to the radiation alone group, which was considered 100%.

Linoleic acid degradation assay

Different concentrations of the extract (500 μ l) were mixed with 5 ml of pre-emulsion (three volumes of linoleic acid + equal volume of Tween-20 in 200 volumes of 30% (v/v) ethanol). Fe²⁺ and Cu²⁺ ions were introduced in the lipid phase using ferrous sulfate (100 μ M), ferric sulfate (100 μ M) and copper sulfate (10 μ M). The total volume was adjusted to 10 ml and exposed to 0.25 kGy gamma radiation (Dose rate: 3.08 kGy/h) followed by incubation at 37 °C for 2 h. Aliquots were taken for assaying the end products of per oxidation using ammonium thiocyanate assay (Kitts et al., 2000).

Statistical analysis

Survival analysis was determined by Kaplan–Meier survival curves and log rank test. For toxicity study Finney’s probit analysis was used. Thymocyte halo assay and endogenous spleen colony were analyzed by mean and standard deviation.

Results

Phytochemical analysis

HPLC analysis of REC-2001 confirmed the presence of 16 compounds. The major compounds identified were podophyllotoxin (1.052%), 4-dimethylpodophyllotoxin glycoside (2.48%), isopodophyllotoxin (0.34%) and podophyllotoxin 1- α - β -D-glucopyranoside (4.35%). Besides, three flavonoids viz., quercetin, kaempferol and kaempferol-3-glucoside were also found to be present in minor traces.

Maximum tolerated dose (MTD) and acute toxicity dose

Single doses of REC-2001 upto 75 mg/kg bwt, were tolerated well by mice without any adverse manifestations. 50% of the animals died within 72 h after administration of 90 mg/kg bwt. All the animals died within 72 h at 115 mg/kg. The predicted model on toxicity data (Table 1) indicated 73 mg as the MTD and 91 mg as LD₅₀ with 95% confidence interval (CI).

Survival studies

All irradiated animals without REC-2001 treatment died within 12 post-irradiation days (Fig. 1). Administration of REC-2001, 10 and 15 mg/kg bwt, 1 h before 10 Gy whole body irradiation rendered > 90% survival upto 30 days. Doses of REC-2001 less than 10 mg/kg or more than 15 mg/kg bwt were ineffective (Fig. 1), but not to the extent of 90% protection. The effect of time interval (0.5, 1, 2, 3 h) between administration of REC-2001 and irradiation was also studied in terms of post-irradiation survival. An interval of 1 h between the administration of REC-2001 and irradiation was found to be optimal for all of the doses of REC-2001 used in the present study (data not shown). The log rank statistics used for survival curve analysis (Table 2) revealed that the dose of 10 and 15 mg indicated a somewhat similar pattern of survival, while the Kaplan Meier survival curves (Fig. 1) showed 15 mg to be a

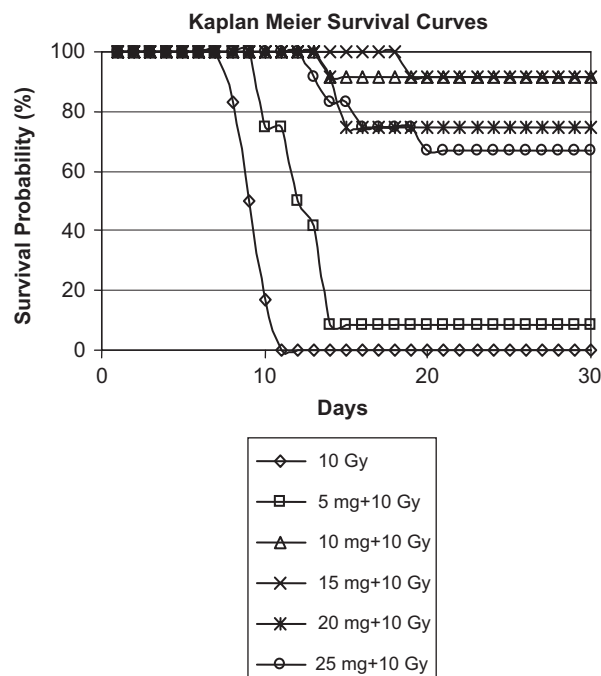


Fig. 1. Different doses of REC-2001 were administered to mice (i.p.) 1 h before exposure to 10 Gy γ -radiation. Survival was observed for 30 days after exposure. Experiments were repeated four times and pooled data was analyzed by Kaplan–Meier survival curves.

more optimum dose reflecting better survival pattern and higher mean survival time (29.08 ± 0.92) with a normal 95% confidence interval.

The relative change in the average weight of animals demonstrated that REC-2001 administration before irradiation rendered minimum reduction in body weight, which also recovered within a short duration in both 10 and 15 mg/kg treated groups as compared to the irradiated group of animals (Fig. 2).

An estimation of DRF on the basis of whole body survival studies against supra-lethal dose (13 Gy) gave an approximate value of DRF = 1.62 (Fig. 3).

Colony forming unit (CFU) assay

Animals given radiation exposure (10 Gy) and REC-2001 in separate groups showed nil CFU counts (Table 3). Pre-irradiation treatment with REC-2001 led to the development of a significant number of colonies on mice spleen ($p < 0.05$). Spleen size was also increased in REC-2001 pretreated-irradiated group, which decreased significantly in radiation control group (gross observation).

Alkaline halo assay

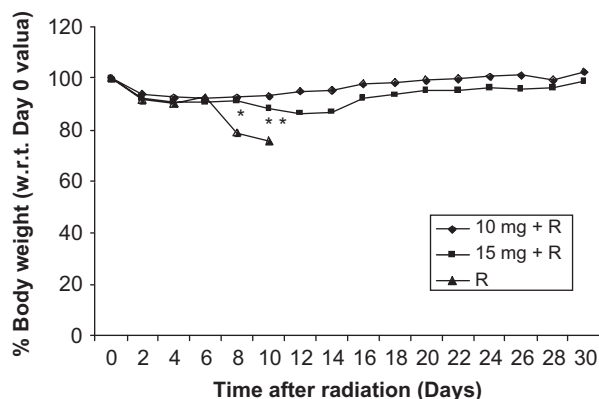
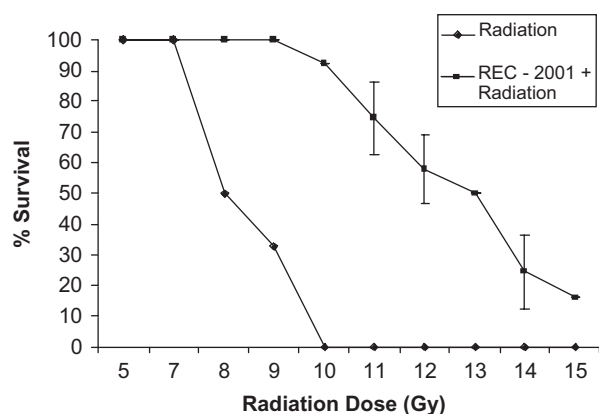
Normal thymocytes, exhibited a halo of diameter $8 \pm 2 \mu\text{m}$ (Figs. 4a and b). Radiation (10 Gy) alone

Table 1. MTD, LD₅₀ and 95% CI of different concentrations of REC-2001 on mice

Prob. of death	REC-2001 (mg/kg bwt)	95% CI
0.05	73	(71.06,75)
0.10	77	(74.24,80.6)
0.25	84	(81.5,87.26)
0.5	91	(85.9,95.12)
0.75	100	(95.76,103.7)
0.90	107	(104,109.6)
0.1	116	(111.8,120.75)

Table 2. Log rank significance study of different doses of REC-2001 (–1 h) before irradiation in male mice

Radiation + REC-2001	10 Gray	10 Gy + 5 mg REC-2001	10 Gy + 10 mg REC-2001	10 Gy + 15 mg REC-2001	10 Gy + 20 mg REC-2001
10 Gy + 5 mg	0.0001				
10 Gy + 10 mg	0.000	0.000			
10 Gy + 15 mg	0.000	0.000	0.9755		
10 Gy + 20 mg	0.000	0.0001	0.3060	0.2579	
10 Gy + 25 mg	0.000	0.0004	0.1437	0.1294	0.6649

**Fig. 2.** *($n = 20$), **($n = 4$). No animal survived in the radiation group beyond day 10 effect of REC-2001 on animal weight in male Strain 'A' mice. Values are expressed as average weight of six treated animals normalized against average weight of six control animals. Data are pooled from four separate sets of experiments.**Fig. 3.** REC-2001 (15 mg/kg bwt) was administered (i.p.; –1 h) prior to 7–14 Gy whole body gamma irradiation. Survival was measured up to 30 days. DRF was estimated as the ratio of the LD₅₀ value of radiation in drug treated group to the un-treated group. Data represented is mean $\pm$ S.D. of four independent experiments.

significantly induced strand breaks resulting in a large halo of diameter $45 \pm 3 \mu\text{m}$ (100% damage). REC-2001 inhibited radiation-induced DNA strand breaks in a

Table 3. Effect of REC-2001 on intrinsic spleen colony forming units (CFU assay) in male Strain 'A' mice

S.No.	Groups	Colony count/spleen (Mean $\pm$ S.D).
1.	Control	0 ± 0
2.	Radiation (10 Gy)	0 ± 0
3.	REC-2001 (10 mg/ kg b wt) + R	5.33 ± 1.15
4.	REC-2001 (15 mg/ kg b wt) + R	7.00 ± 1.00

Animals from two groups were administered REC-2001 (10 and 15 mg/kg bwt, –1 h) before 10 Gy gamma irradiation. Visible colonies on spleen surface were counted. Values are average of counts from six spleens per group.

concentration dependent fashion and at a concentration of $500 \mu\text{g/ml}$, it almost completely blocked strand breaks and the halo diameter found to be $15 \pm 2 \mu\text{m}$ (35.5%).

Linoleic acid degradation assay

The ability of the different concentrations of extract to restrict Fe^{2+} , and Cu^{2+} (in combination with 0.25 kGy)-induced lipid oxidation is depicted in Figs. 5a and b. Maximum inhibition was found at concentrations of 100 and $30 \mu\text{g/ml}$, respectively. A slight pro-oxidant activity was observed at $\log [\text{extract}] = 0.477$ in all the cases of stress induction.

Discussion

Exposure to lethal doses of ionizing radiation severely increases the oxidative burden on the body and endogenous antioxidant defense mechanisms eventually cannot cope with this increased stress. Therefore, in such situations extraneous antioxidants have been put into trials to prevent lethal damage (Shimoi et al., 1994; Weiss and Landauer, 2003). A continuous search for effective and non-toxic radio-protectants by our group has revealed the radioprotective potential of high altitude *Podophyllum hexandrum*. (Goel et al., 2003;

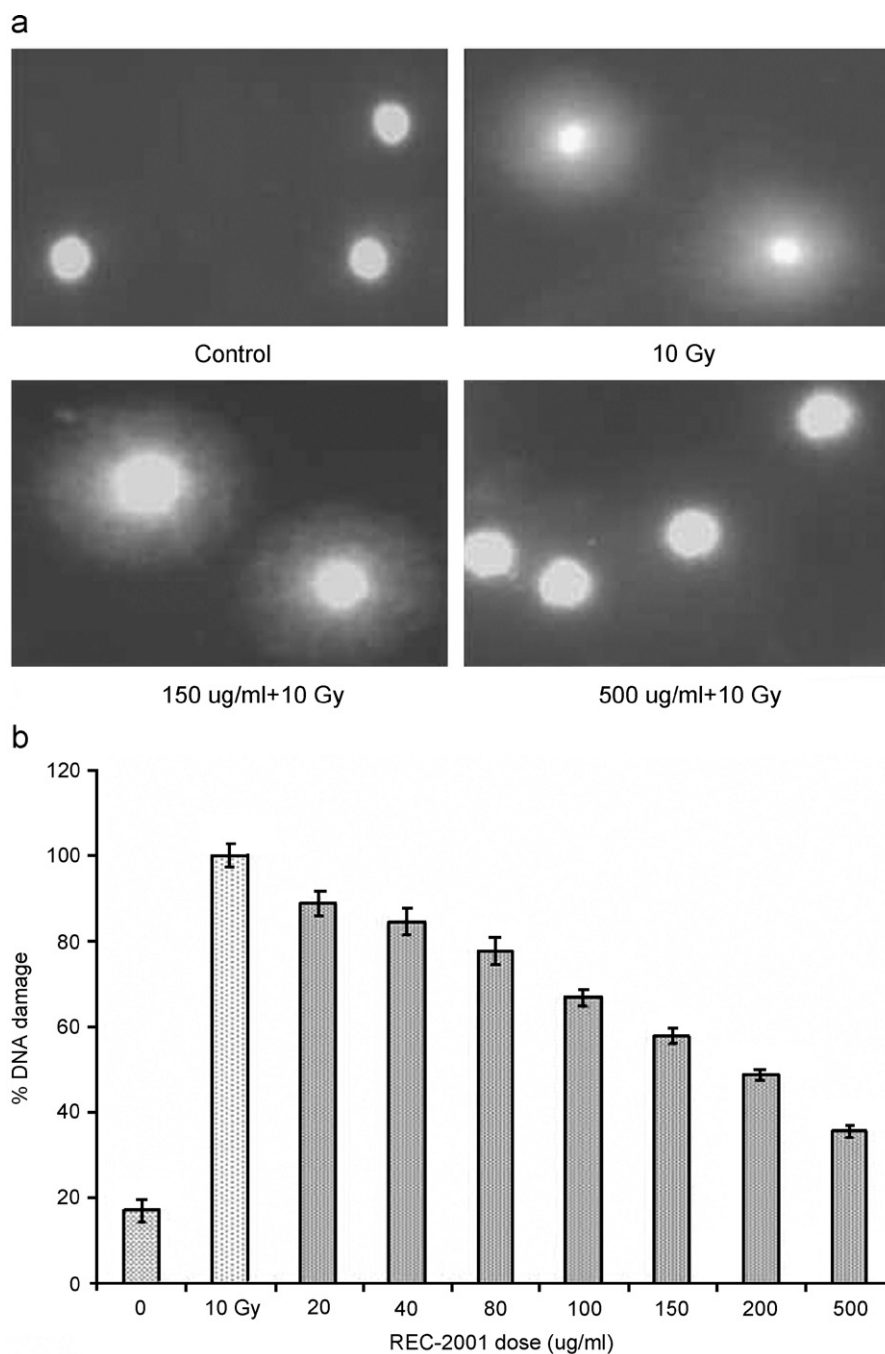


Fig. 4. (a) Effect of REC-2001 on DNA fragmentation in thymocytes of male Strain ‘A’ mouse. Thymocytes isolated from normal mouse were divided in three groups, viz., control (non-irradiated), radiation exposed (10 Gy) and REC-2001 treated (different doses) before radiation exposure. (b) Inhibition of percentage DNA fragmentation by REC-2001. Halo radius was measured from the center to any point on the perimeter. Data points are mean values of 100 cells from each group.

Arora et al., 2005; Chawla et al., 2005). The use of whole extract of *Podophyllum hexandrum* in human beings has been reported to be toxic if used in larger concentrations, mainly due to the presence of podophyllotoxin (Chang et al., 1992). From REC-2001, a major amount of podophyllotoxin (except 1.05%) was removed. HPLC analysis done as a part of this study and supporting

LC-APC I-MS/MS data however, have confirmed the presence of other lignans (Sagar et al., 2006). Whole-body survival to the level of >90% in pre-treated (REC-2001) irradiated group against 0% survival in radiation control group has revealed that less quantity of podophyllotoxin in REC-2001 has not decreased the radioprotective efficacy of the extract.

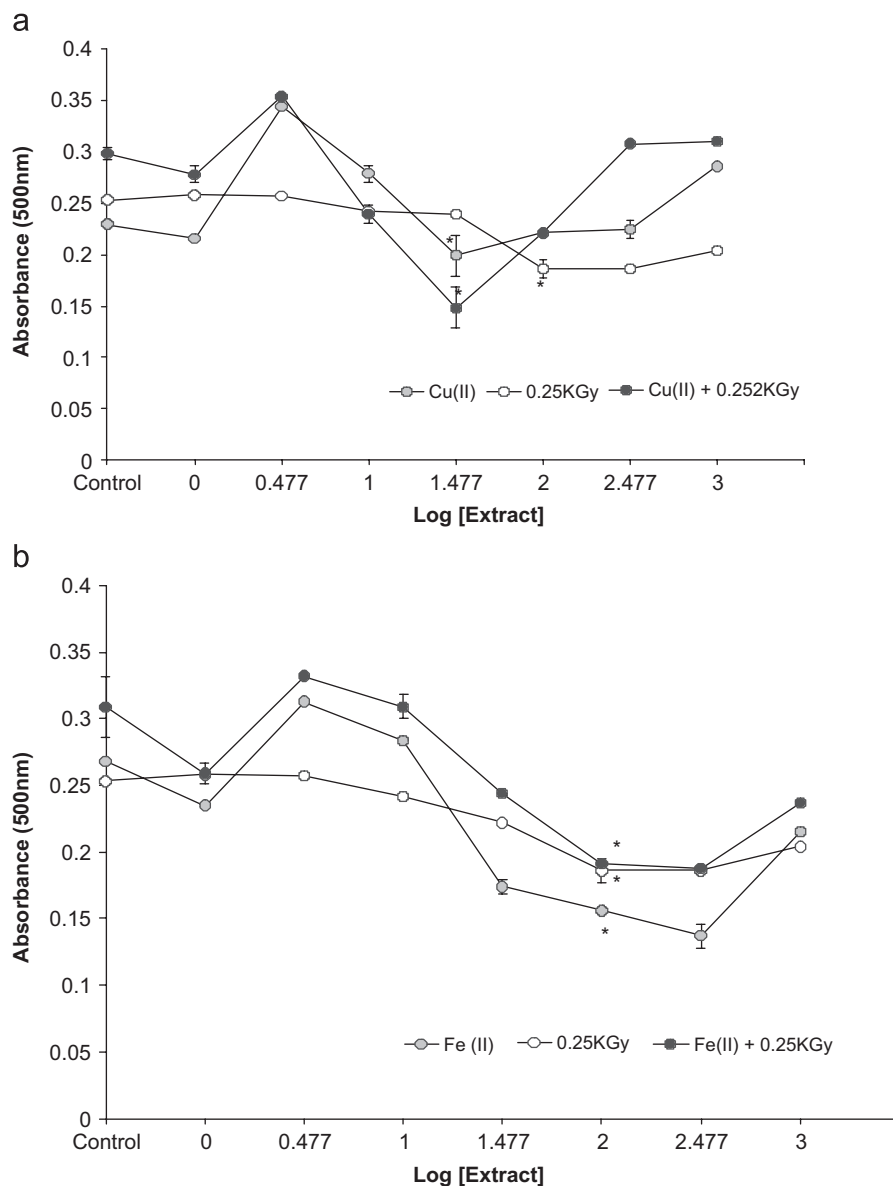


Fig. 5. Effect of REC-2001 on linoleic acid degradation assay. Different concentrations of REC-2001 were mixed with 5 ml of pre-emulsion. Fe^{2+} ions were introduced in the lipid phase using $100\text{ }\mu\text{M}$ Fe_2SO_4 and $100\text{ }\mu\text{M}$ Fe_3SO_4 (a), whereas Cu^{2+} ions were introduced in the lipid phase using $10\text{ }\mu\text{M}$ CuSO_4 (b). The total volume was exposed to 0.25 kGy gamma radiation (Dose rate: 3.08 kGy/h).

Significant difference between MTD (75 mg/kg bwt) and effective doses ($10\text{--}15\text{ mg/kg bwt}$) supported the effort of reducing toxicity without compromising with survival, as a wise step. Most effective time gap of 1 h between extract administration and irradiation has indicated that major bioactive molecules were available in optimal concentration in the cellular environment at 1 h. Animals treated with the most effective doses of REC-2001 rendered significant protection against radiation-induced severe loss in body weight (Fig. 2), which indirectly indicates that gastrointestinal tract and other vital organs too have received significant protection in irradiated mice by REC-2001.

The higher proliferative activity of haemopoietic stem cells (CFU assay, Table 3) in the extract-treated group revealed an immunostimulatory property of REC-2001, and the same was reported earlier with the crude extract of this plant (Goel et al., 2003). Phytoextracts are well known for cellular DNA protection properties (Jagetia and Aruna, 1997; Abalea et al., 1999) mainly by reducing oxidative stress, chelating transition metal ions, reducing lipid peroxidation and enhancing endogenous defense mechanisms. Significant decrease in the diameter of mice thymocyte halo in treated irradiated group (Figs. 4a, b) strongly supports the view of DNA protection rendered by REC-2001. Monitoring

of radiation-induced membrane lipid peroxidation and its modulation by REC-2001 (Figs. 5a, b) indicated that this extract possesses the ability to reduce the amplifying species even from the lipid system. These results were consistent with our earlier findings using high altitude *Podophyllum hexandrum* against radiation-induced lipid peroxidation (Chawla et al., 2005). Most of the synthetic (Sweeney, 1979) and some herbal radioprotective (Goel et al., 2002) agents often render maximum radioprotective effect at the doses approaching MTD. However, REC-2001 exhibited significant radioprotection at much lower concentrations than its MTD (75 mg/kg). Its effective dose was also found to be 9–10 times less than the lethal dose. The alcoholic fraction of *Podophyllum hexandrum* is, therefore, likely to find greater clinical acceptance for human use since it has a greater therapeutic index. The DRF of 1.62 further supports the view of using of REC-2001 against supra-lethal doses of radiation.

Phytochemical analysis of this fraction has enabled us to confirm the role of lignans and flavonoids in rendering protection through various modes of action. However, it is difficult to presently comment on efficacy of a particular molecule in REC-2001 playing a lead role in rendering whole-body protection against lethal radiation. Polyphenols and lignans present in REC-2001 when tried in purified forms (unpublished data) never performed significantly against lethal radiation. In fact polyphenols in the presence of lignans and other compounds further enhance their protective efficacy by acting in synergism and antagonism and generating the capability to terminate radical chain reactions and enhancing antioxidant defense mechanism. However, further fractionation of REC-2001 and current efforts of our group to unravel mechanistic aspects of REC-2001 might explain the efficacy of individual molecules against radiation at some future time.

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